AI Critique

The AI was useful for broad coverage, but the audit shows that its first-pass tests were often not execution-ready. FR-05 initially had 10 INCOMPLETE AI cases, mainly because inputs such as “long string,” SQL-oriented payloads, or error triggers were descriptive rather than deterministic. FR-10 had six INCOMPLETE cases for similar reproducibility and evidence issues, while FR-16 had seven before correction, with AI-FR16-046 remaining blocked because no safe deterministic database-error trigger existed. No AI-generated testcase was ultimately classified INVALID, so the main weakness was incompleteness rather than wholesale irrelevance.

The clearest AI blind spot was compositional and cross-endpoint reasoning. Human-added FR-05 cases combined parser edge conditions and write/read sequences that the AI had not proposed. HUMAN-FR05-044 used an encoded null byte and exposed the SEC-05 SQL-construction defect. HUMAN-FR05-048 combined an unauthenticated malformed POST with before/after verification and revealed both missing authorization and invalid all-null product persistence. Human cases also added rename→search, delete→search, duplicate JSON keys, method override, JWT alg:none, concurrency, stale category references, and repeated invalid-batch behavior. These gaps occurred because the generation prompts were primarily scoped around one endpoint and approved requirement partitions, encouraging systematic local coverage but less temporal, parser-layer, and cross-feature exploration.

However, AI did identify the FR-10 shipping-state cancellation defect and the main FR-16 price, rollback, and role-authorization failures once deterministic runtime fixtures existed. The lesson is that AI should generate breadth, while humans challenge assumptions, add cross-layer scenarios, make vague tests reproducible, and decide whether runtime failures truly violate the contract.